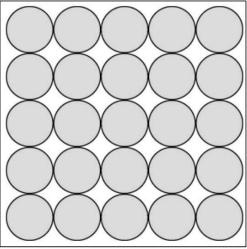


2.1 Kinetic Particle model of matter

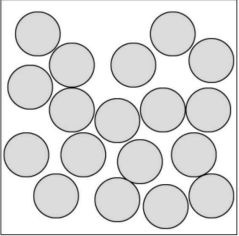
States of matter

- Matter is made up of particles: atoms, molecules, ions, electrons

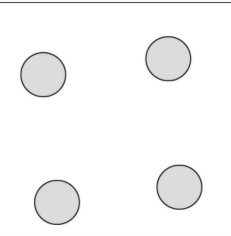
Solids:

	- Intermolecular forces between particles are very strong -> So, particles are very close together, and arranged in a regular pattern - Particles cannot move freely -> definite shape - Particles are not easily compressible -> definite volume - Particles can only vibrate in fixed positions
---	--

Liquids:

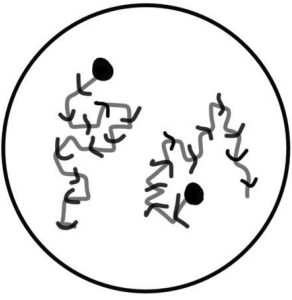
	- Intermolecular forces between particles are weaker than in a solid, but still strong -> So, particles are close together, but arranged in an irregular pattern - Particles can move freely and flow to take shape of container -> no definite shape - Particles are not easily compressible -> definite volume - Particles slide past each other
---	---

Gases:

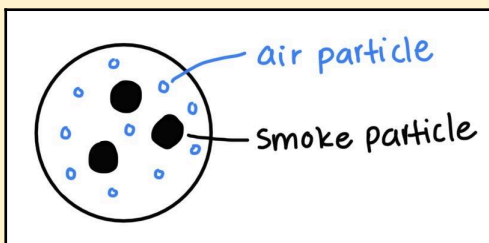
	- Intermolecular forces between particles are very weak -> So, particles are far apart, and are arranged in an irregular pattern - Particles can move freely and flow to take shape of the container -> no definite shape - Particles are easily compressible -> no definite volume - Particles move in random directions at high speeds
---	---

Brownian Motion

- The random motion of microscopic particles suspended in a liquid or gas
 - It is evidence that matter is made out of tiny particles in constant motion (Kinetic particle model of matter)
-
- Small particles (such as smoke or pollen) are suspended in a liquid or gas
 - They are observed through a microscope

	Describe: - Microscopic particles are seen constantly moving and changing directions, in zig zag patterns
--	--

Explanation:



- Smoke particle is much larger than air particle, which is why we can see the smoke particles but not air particles through the microscope
 - Air particles move in random directions at high speeds around the smoke particles.
 - Air particles collide with smoke particles in all directions, causing the smoke particles to be seen constantly moving and changing directions, in zig zag patterns
-

Absolute scale of temperature

Absolute zero: the lowest possible temperature, where particles have no kinetic energy

- Absolute zero is 0K, which is equal to -273°C

$$\text{Temperature (in Kelvin)} = \theta \text{ (in Celsius)} + 273$$

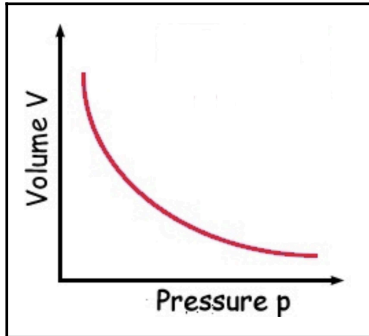
Pressure of a gas:

- Pressure of a gas is produced by the collisions of the gas molecules on the surface of an object
- When gas molecules collide with surfaces, they bounce off, causing a change in their velocity direction
- Momentum = mass x velocity, so there is also change in the momentum of gas molecules
- Force = change in momentum / time, so collisions result in force being exerted in the surface of an object
- So, there is pressure, as pressure is the force per unit area

Pressure and Volume - Boyle's law

- When mass and temperature of a gas remains constant, pressure is inversely proportional to volume

Graph:



Formulas:

$$p = \frac{k}{V} \quad k = pV$$

$$p_1 V_1 = p_2 V_2$$

When volume increases:

- When volume of container increases, gas molecules are further apart
- This causes gas molecules to collide with walls of the container less frequently
- Total force per unit area decreases, so pressure decreases

When volume decreases:

- When volume of container decreases, gas molecules are closer together
 - This causes gas molecules to collide with the walls of the container more frequently
 - Total force per unit area increases, so pressure increases
-

Pressure and Temperature:

- When mass and volume of a gas remains constant, pressure is directly proportional to temperature

When temperature increases:

- When temperature increases, gas molecules gain kinetic energy and move faster.
- Gas molecules collide with the walls of the container more frequently
- Total force per unit area increases, so pressure increases

When temperature decreases:

- When temperature decreases, gas molecules lose kinetic energy and move slower
- Gas molecules collide with the walls of the container more frequently
- Total force per unit area increases, so pressure increases

2.2 Thermal properties and temperature

Thermal expansion:

- When a solid, liquid or gas is heated,
- Particles gain kinetic energy, and vibrate or move faster
- This causes particles to move further apart, increasing distance between particles, which causes volume of substance to increase

Thermal contraction:

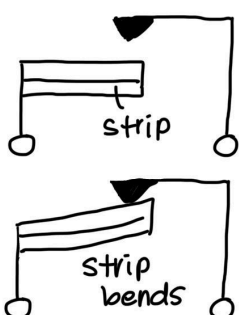
- When a solid, liquid or gas is cooled,
- Particles lose kinetic energy, and vibrate or move slower
- This causes particles to move closer together, decreasing distance between particles, which causes volume of substance to decrease

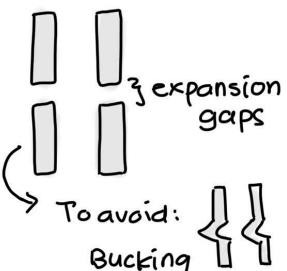
Comparing solid, liquid and gas:

- Solids have very strong intermolecular forces between particles -> so, difficult to expand and contract
- Liquids have strong intermolecular forces between particles, but not as strong as in solids -> so, easier to expand and contract compared to solids
- Gases have very weak intermolecular forces between particles -> so, easy to expand and contract

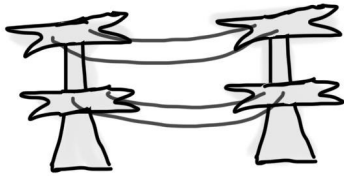
- In conclusion, Gases > Liquids > Solids (to expand/contract)

Everyday applications of thermal expansion

	Bimetallic strip: - The strip is made up of 2 different metals that expand at different rates when heated - The bottom metal expands faster than the top metal, causing it to bend upwards - This is used in temperature activated switches
--	--

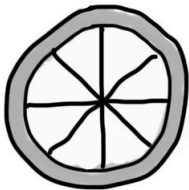
	Railway lines: - There are expansion gaps to allow for thermal expansion - This prevents the tracks from bucking when it gets hot during the summer
---	--

	Thermometer: - When heated, the liquid expands and rises up the tube - When cooled, the liquid contracts and goes down the tube
---	--



Electric cables:

- Cables have slack to allow for thermal expansion and contraction
- This prevents the cables from breaking when it is too cold



Car tires:

- Air in car tires expand when heated and contract when cooled
- So, car tires are inflated to a higher pressure in the summer compared to the winter, so it is harder for the air to expand

Specific heat capacity

Specific heat capacity: the energy required to increase 1 unit mass of a substance by 1 unit temperature

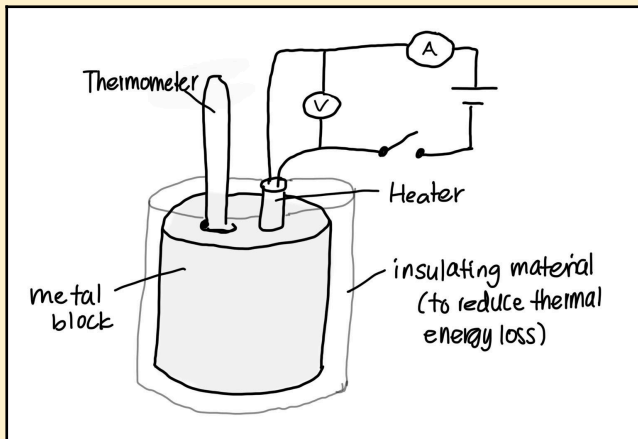
$$Q = mc\Delta T$$

Diagram illustrating the equation $Q = mc\Delta T$ with labels for each term:

- Q : energy (J)
- m : mass (kg)
- c : specific heat capacity ($\text{J/kg}^\circ\text{C}$)
- ΔT : Temperature ($^\circ\text{C}$)

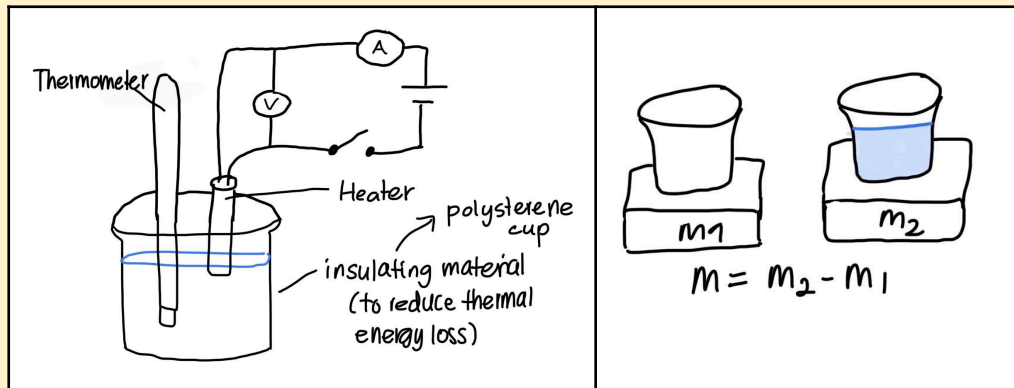
- Specific heat capacity remains constant for a given substance

Experiment - to measure specific heat capacity (metal block)



- Drill 2 holes into the metal block
- Using a balance, measure the mass of the block. Record as m .
- Set up apparatus as shown in the diagram
- Place thermometer in one hole, and heater in the other. Connect the heater to a power supply, switch, voltmeter and ammeter.
- Measure initial temperature. Record as T_1
- Switch on current, and immediately start stopwatch.
- Measure current using ammeter, record as I
- Measure voltage using voltmeter, record as V
- After 60 seconds, measure final temperature. Record as T_2
- Calculate temperature change: ($\Delta T = T_2 - T_1$)
- Calculate energy: ($E = VIt = VI (60)$)
- Calculate specific heat capacity: ($c = E / (m\Delta T)$)

Experiment - to measure specific heat capacity (liquid)



- Measure mass of liquid using electronic balance. Record as m . (mass of cup with water - mass of cup alone)
- Set up apparatus as shown in the diagram
- Place a thermometer and heater into the beaker. Connect the heater to a power supply, switch, ammeter and voltmeter
- Measure initial temperature. Record as T_1
- Switch on current, and immediately start stopwatch.
- Measure current using ammeter, record as I
- Measure voltage using voltmeter, record as V
- After 60 seconds, measure final temperature. Record as T_2
- Calculate temperature change: ($\Delta T = T_2 - T_1$)
- Calculate energy: ($E = VIt = VI (60)$)
- Calculate specific heat capacity: ($c = E / (m\Delta T)$)

- The Specific heat capacity in this experiment which is calculated, will be higher value than actual
 - This is because some thermal energy is lost to the surroundings, causing a smaller change in temperature.
-

Boiling and evaporation

Boiling

- The liquid reaches its boiling point
- It continues to absorb energy to overcome the intermolecular forces and separate the particles, turning into the gas

Evaporation:

- The most energetic particles at the surface of the liquid overcomes the intermolecular forces
- The particles separate and they escape from the liquid surface into the air as gas
- The lower energy particles are left behind, which causes temperature of the liquid to decrease

Similarity and differences between boiling and evaporation

Similarity:

- Both changes state from liquid to gas

Differences:

- Evaporation takes place at the surface of the liquid, while boiling takes place throughout the liquid
 - Evaporation occurs at any temperature between melting and boiling point, while boiling occurs at boiling point only
 - For evaporation, temperature decreases. For boiling, temperature remains constant
-

Factors that affect rate of evaporation

Temperature:

- The higher the temperature, the faster the rate of evaporation
- There are a greater proportion of particles which have enough energy to overcome intermolecular forces, and escape into air as gas

Surface area:

- The larger the surface area, the faster the rate of evaporation
- There are more particles at the surface of the liquid, so more highest energy particles are able to escape into the air as gas

Wind speed:

- The faster the wind speed, the faster the rate of evaporation
- The wind removes water vapour from the surface of the liquid, so more highest energy particles are able to escape into the air as gas

Humidity:

- The lower the humidity, the faster the rate of evaporation
 - There is less water vapour in air, so more highest energy particles are able to escape into the air as gas
-

2.3 Thermal energy transfers

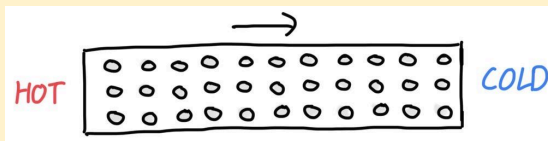
Thermal energy transfers:

- If there is a difference in temperature, thermal energy is transferred from the hotter object to the colder one
 - This will continue until both objects are at the same temperature, meaning they are in thermal equilibrium
 - Thermal energy is transferred by -> conduction, convection and radiation
-

Conduction

- Thermal energy is transferred through the vibration of particles
- Occurs mainly in solids
- Not in liquids and gases, because particles are free to move around -> they instead transfer energy through convection

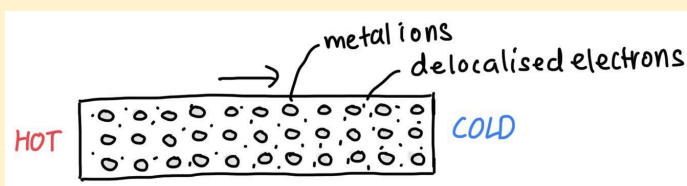
Describing conduction: (non-metal)



- At the hot end of the bar, particles gain kinetic energy and vibrate faster
- Particles collide with neighbouring particles, which also start to vibrate faster
- Thermal energy is transferred from the hot end to the cold end of the bar

- For metal, there are delocalised electrons, so thermal energy transfer by conduction is faster

Describing conduction: (metal)



- At the hot end of the bar, metal ions gain kinetic energy and vibrate faster, and delocalised electrons also gain kinetic energy and move faster
- Metal ions and delocalised electrons collide with other metal ions and electrons
- Thermal energy is transferred from the hot end to the cold end of the bar

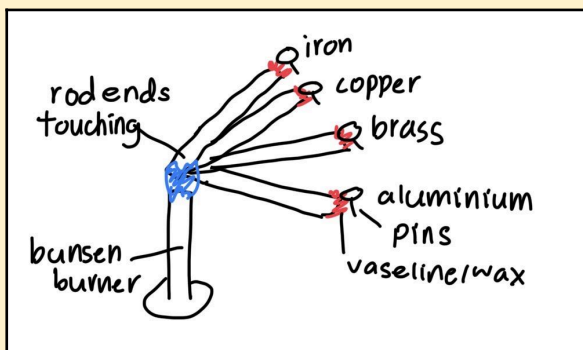
Good conductors:

- Substances that can transfer thermal energy easily, because they have free moving electrons

Poor conductors: (insulators)

- Substances that cannot transfer thermal energy easily, because they do not have free moving electrons
- Gases are poor conductors because particles are far apart, so energy cannot be transferred from one particle to another

Experiment - to distinguish between a good and bad thermal conductor



- Use 4 metal rods
- Attach pins to the end of each rod, using 2cm³ vaseline
- The other end of the rods without vaseline is brought together. Heat this end with a bunsen burner.
- Immediately start stopwatch, and record the time taken for each rod's pin to fall off
- The best conductor is the rod which the pin falls off in the shortest time.

*for P6, IV would be the material of the rod. So, do 1 material rod first, then repeat using other metal rods of different material

Experiment - to show that water is a poor conductor of heat

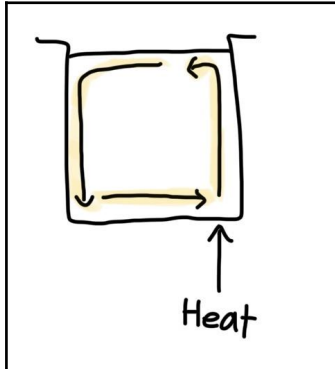


- Trap ice at the bottom of a test tube, with a piece of metal gauze
- Heat the water at the top of the test tube until it boils
- Ice at the bottom does not melt, so water is a poor conductor of heat.

- Note: if ice is allowed to float normally and the water at the bottom of the test tube is heated, it melts quickly -> this is because water molecules can move, and water heats by convection.
 - Water, like other non-metals is a poor conductor of heat because it does not have free moving electrons to transfer thermal energy
-

Convection

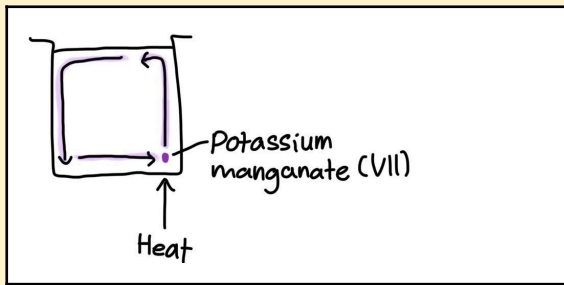
- The transfer of thermal energy by the movement of currents and particles within a fluid (liquid and gas)



Describing convection:

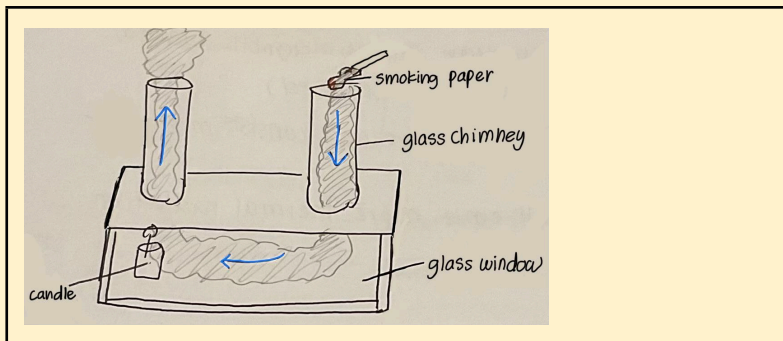
- The water is heated. Particles gain kinetic energy and move faster.
- They move further apart, causing volume to increase and density to decrease
- The warm, less dense water rises
- For the cooler water, the particles have less kinetic energy and move slower.
- They move closer together, causing volume to decrease and density to increase
- The cool, more dense water sinks.
- The cool water then replaces the warm water, and it is heated.
- This process repeats, forming a convection current.
- The process repeats until water is heated evenly throughout

Experiment - to demonstrate convection: (liquids)



- Place potassium manganate (VII) crystals at the bottom of a beaker filled with water
- Heat the beaker directly below the crystals using a Bunsen burner
- Observe the movement of the purple dye. The heated water will become less dense, rise and carry the dye upwards. Then it will cool, become more dense and sink, carrying dye downwards.
- The movement of the purple dye is the convection current

Experiment - to demonstrate convection: (air - glass chimney)



- Set up box and chimney as shown in the diagram
- Light a candle and place it under one of the glass chimneys
- Heat smoking paper and place it above the other chimney
- Observe the movement of the smoke.
- The smoke gets sucked down instead of rising
- Explanation: the air heated by the candle becomes less dense and rises. The cooler air from the other chimney sinks and moves to replace this air. The smoke is moved with it

Materials with trapped air:

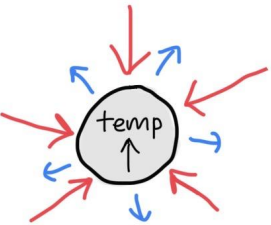
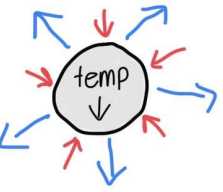
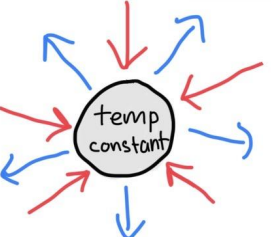
- Examples: cotton, wool and bubble wrap
 - Materials that have trapped air are good insulators.
- Air does not conduct heat
 - Convection cannot take place in trapped air
-

Radiation

- The transfer of thermal energy through electromagnetic waves
 - Can occur in a vacuum (particles are not involved)
 - Emitted by all bodies above absolute zero, and consists of mostly infrared radiation
- As object gets hotter, it will emit more thermal radiation

Thermal equilibrium:

- When objects absorb more energy, it gets hotter

	- The rate at which an object emits energy is less than the rate at which it absorbs energy - Temperature increases
	- The rate at which an object absorbs energy is less than the rate at which it emits energy - Temperature decreases
	- The rate at which an object absorbs energy is equal to the rate at which it emits energy - Temperature remains constant (thermal equilibrium)

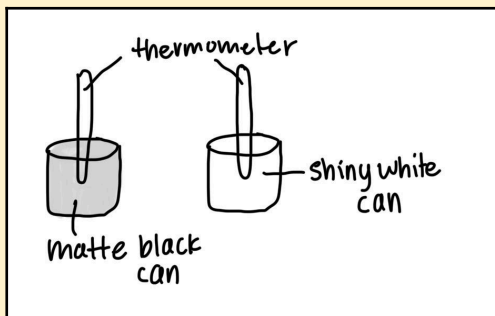
Emitters and absorbers:

- Matte / dull black surface = good emitter and good absorber
- Shiny white surface = good reflector, bad absorber and bad emitter

Factors affecting rate of emitted energy:

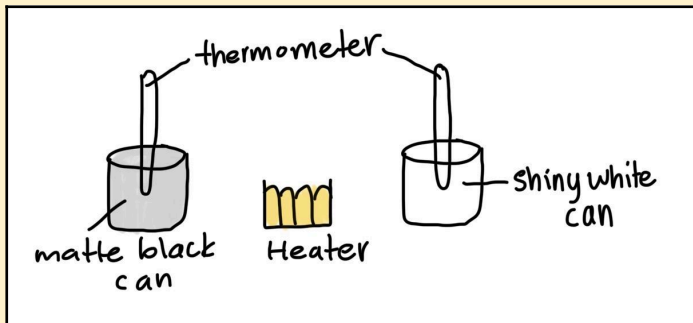
- Temperature: the higher the temperature, the faster the rate at which the object emits energy
- Surface area: the larger the surface area, the faster the rate at which the object emits energy

Experiment - to distinguish between good and bad emitters



- Fill one matte black can and one shiny white can with 50cm^3 of hot water.
- Measure initial temperature using a thermometer. Record as T_1
- After 2 minutes, measure the final temperature. Record as T_2
- Calculate the temperature decrease ($T_1 - T_2$)
- Can with larger temperature decrease is a better emitter

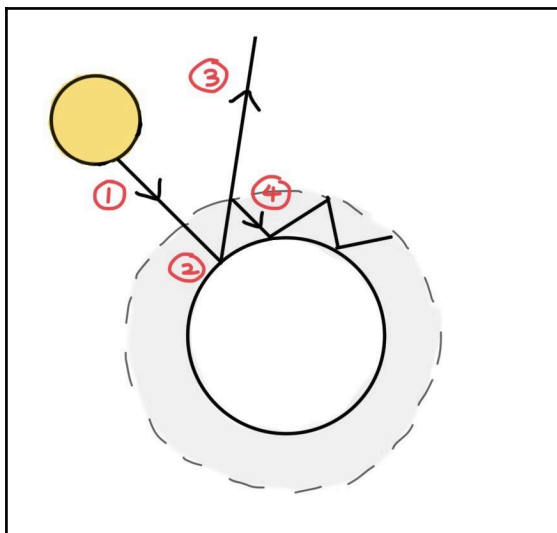
Experiment - to distinguish between good and bad absorbers



- Place one matte black can and one shiny white can 20cm away from a 30W heater
- Fill the 2 cans with 50cm^3 of 25°C water
- After 5 minutes, measure the final temperature using a thermometer. Record as T
- Calculate the temperature increase ($T-25$)
- Can with larger temperature increase is a better absorber

Greenhouse effect

- The atmosphere of the earth is what causes the earth's temperature
- If Earth have no atmosphere, the earth's surface would emit all the radiation from the sun into space



Greenhouse effect:

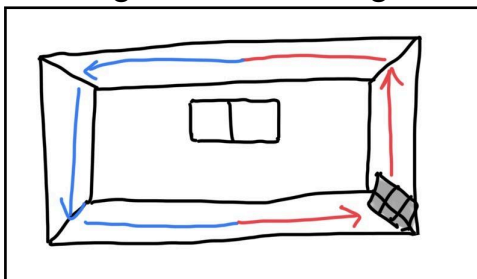
- Radiation from the sun enters the earth's atmosphere to reach the earth's surface
- Radiation is absorbed by the earth's surface, then re-emitted at longer wavelengths
- Some radiation passes through atmosphere into space
- Some radiation is absorbed by greenhouse gases and is re-emitted back to earth's surface. This increases the temperature of the earth's surface

So, the temperature of the Earth is affected by several factors:

- Rate of radiation from sun reflected back into space
 - Rate of radiation absorbed by the earth's atmosphere and surface
 - Rate of radiation emitted from the earth's surface and atmosphere into space
-

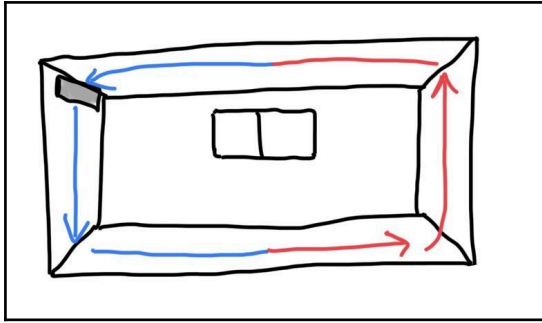
Consequences of thermal energy transfer

Heating the room using a heater:



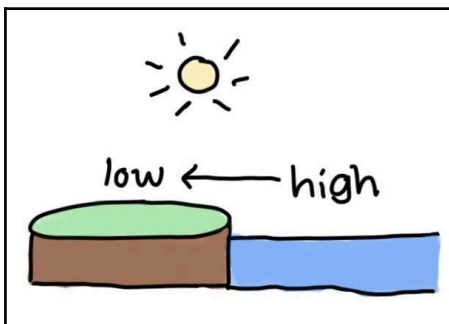
- The air above the heater is heated. It becomes warmer, less dense and rises
- Cooler air is more dense, and sinks
- The Cool air flows to replace the warm air, then it is heated
- This process repeats, forming a convection current
- The process repeats until the room is heated evenly

Cooling the room using air conditioner:



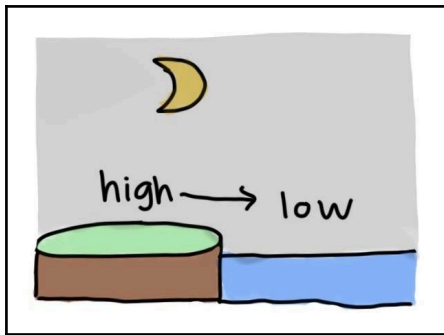
- The air below the air conditioner is cooled. It becomes cooler, more dense and sinks
 - Warmer air is less dense and rises
 - Warm air flows to replace the cold air, then it is cooled
 - This process repeats, forming a convection current
 - The process repeats until the room is cooled evenly
-

Sea breezes:



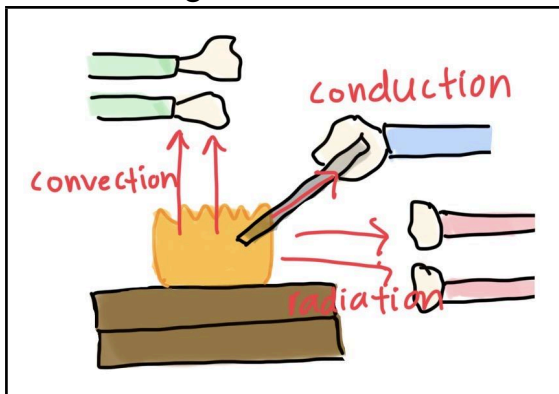
- During the day, the sun heats the land up more than the sea
- Air above the land becomes warmer, less dense and rises, creating a low pressure area
- Air above the sea is cooler, more dense and sinks, creating a high pressure area
- The cooler air above the sea flows to the land, to fill in the low pressure. This is a sea breeze.

Land breezes:



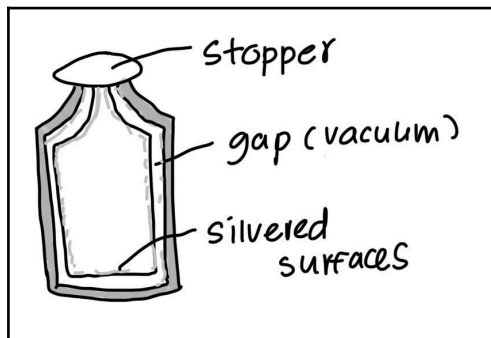
- During the night, the land cools more quickly than the sea.
- Air above the land becomes cooler, more dense and sinks, creating a high pressure area
- Air above the sea is warmer, less dense and rises, creating a low pressure area
- The cooler air above land flows to the sea, to fill in the low pressure area. This is a land breeze.

Fire burning wood:



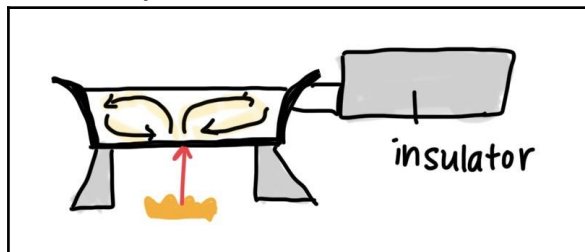
- Thermal energy is transferred through the solid to the hand by conduction
- Thermal energy is transferred through the air to the hands at the side of the fire by radiation
- Thermal energy is transferred through the air to the hands above the fire by convection -> because air above a fire is heated, become less dense and rises

Vacuum flask:



- The stopper is made out of insulator, to reduce thermal energy loss by conduction
- Gap contains no air (vacuum), so that there are no particles to transfer thermal energy by conduction or convection
- The inner sides of the flask are silver, to reduce thermal energy loss by radiation

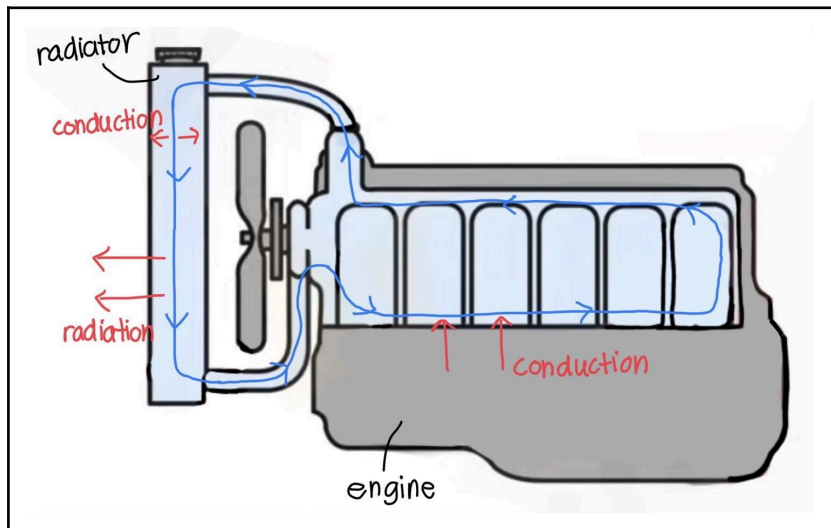
Kitchen pan:



- Thermal energy from the fire is transferred to the metal pan by convection and radiation
 - Thermal energy is transferred through the metal pan by conduction
 - Thermal energy is transferred from the metal pan to the water by conduction
 - Thermal energy is transferred through the water by convection
 - Plastic handle is an insulator, to reduce thermal energy loss by conduction through the handle
-

Radiator in a car:

- Role is to transfer thermal energy away from the engine, as the engine reaches high temperatures



- When liquid passes over engine, thermal energy is transferred from the engine to the liquid by conduction
 - This liquid travels back to the radiator, and transfers thermal energy to the radiator, by conduction
 - The radiator then transfers thermal energy to the surroundings by radiation
-